

JUNIOR SCHOOL EDUCATION

GRADE 9

AGRICULTURE AND NUTRITION

NOTES

JUNIOR SECONDARY GRADE 9 AGRICULTURE & NUTRITION COMPLETE RATIONALISED NOTES

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STRANDS COVERED IN THE GRADE

**Conservation of Resources.
Food Production Process.
Hygiene practices.
Production Techniques.**

Strand1: CONSERVATION OF RESOURCES.

1.1 Conserving of Animal Feeds: Hay.

Meaning of Conservation.

- ✓ Drought is a period of prolonged dry weather.
- ✓ Drought reduces quantity and quantity of forage. Due to this it is important to conserve forage for future use during dry periods.
- ✓ **Conservation** refers to efficient use of scarce resources through reduction of loss to minimize waste.
- ✓ Conservation therefore, requires one to make correct decisions and choices to ensure there is enough for the future.

Ways of Conserving resources include:

- ✓ Conserving of animal feeds.
- ✓ Conserving of leftover foods.
- ✓ Practicing integrated farming.

Conserving of animal feeds: Hay

- ✓ **Forage** refers to plants which grow naturally or are planted by man which are used as livestock feed or eaten by domestic animals.
- ✓ These plants can be conserved and used when there is scarcity.
- ✓ Examples of forage include: grasses, legumes or a mixture of grasses and legumes.
- ✓ Forage is either given directly to animals through grazing or it can be harvested at a certain stage and fed to animals.
- ✓ Conserving forage ensures there is enough during drought.
- ✓ Forage can be conserved in form of hay.
- ✓ **Hay** refers to grass, legumes or other plants that are cut, dried and stored to be used by livestock as feed.
- ✓ Suitable crops for making hay include:
 - × Grass.
 - × Maize stover.
 - × Napier grass.
 - × Rhodes grass.
 - × Wheat straw
 - × Desmodium.
 - × Sorghum .
 - × Baley
 - × Oats etc.

Grass 	Maize stover 	Napier grass 	Rhodes grass. 
Wheat straw 	Desmodium 	Sorghum. 	

Methods of Conserving Forage.

✓ Forage can be conserved using three methods as follows:

- Baled hay. (baling)
- Standing forage.
- Stacked hay. (Creating haystacks)

Baled hay and Stacked hay (baling)?



Baled hay



Stacked hay

- ✓ Baled hay is dried forage that has been compacted in special structures to form bales which are stored for a long time.
- ✓ Hay making process involves cutting, spreading cut forage for faster drying and putting it in rows (**windrowing**) for easier collection.
- ✓ The structures that can be used to make bales include:
 - ✗ Metallic or wooden Box baler.
 - ✗ Wooden box



Materials required to make baled hay:

- ✧ Dried forage such as grass, maize stover or straw.
- ✧ Metallic or wooden Box baler/wooden box
- ✧ Sisal twine/strings.
- ✧ Gumboots.
- ✧ Gloves.
- ✧ Panga/sickle.
- ✧ Knife.
- ✧ Rake.

Procedure followed when making baled hay.

- ☆ Wear gloves and gumboots to protect hands and feet.
- ☆ Cut fresh forage from the field using either a panga or a sickle.
- ☆ Spread to allow the cut forage to dry for 2 to 3 days using a rake.
- ☆ You can also use already dried forage materials from the field such as maize stover or straw or grass.
- ☆ Put the dried forage materials into the box baler or wooden box such as 40cm x 50cm x 75cm.
- ☆ Put s lengths of sisal twine in the box
- ☆ Compact the materials by pressing it down tightly or applying pressure from the top.
- ☆ Tie the materials securely or bind the compressed hay using binding materials such as sisal twine, strings etc.
- ☆ Remove baled hay from the box
- ☆ Store the hay bales in a dry shaded area for use during drought.

Advantages of making hay bales.

- ✗ They are easy to transport.
- ✗ Bales offer easy and safe storage for a long time.
- ✗ It is easy to handle hay while it is in form of bales.
- ✗ Bales requires less space.

Disadvantages of bailing.

- ✧ It is laborious/ a lot of work is required.
- ✧ Depends on weather.

Locally available materials you can use to make baled hay.

- A stiff Carton box or wooden box.
- Rhodes grass or straw.
- Sisal twine /any tying materials.
- Gloves.
- Gumboots.

Reasons why baling should take place when it is completely dry.

Baling should take place when it is completely dry to;

- prevent rotting and mould growth.
- Reduce spoilage.
- Preserve nutritional quality.

Types of bales that can be made during baling process in making of hay:

Bales can be:

- ✧ Rectangular.
- ✧ Square.
- ✧ Cylindrical.

Activities involved in making baled hay:

- ✧ Cutting of forage.
- ✧ Spreading of cut forage in the sun for easy and faster drying.
- ✧ Compressing of forage into bundles(bales) of forage materials into compact units called bales.

Stacked hay.

Staking hay refers to piling of dried forage in a heap. Materials like maize stovers or wheat straws are cut and piled on a raised surface.

The forage should be kept in a well sheltered place to prevent rainwater and sunlight.



Materials required when making haystacks (Stacking)

- ✧ Dried forage materials such as grass, maize stover, Rhodes grass or straw, oats, barley,.
- ✧ Sturdy pieces of timber or metal.
- ✧ Posts, rafters, nails, hammer, wood saw, crowbar, tape measures and panga
- ✧ Gloves.
- ✧ Gumboots.

Advantages of haystacks.

- ☆ Easy to assemble.
- ☆ Suitable in areas without equipment for making baled hay.

Disadvantages of haystacks.

- Require large spaces.

Factors to consider when choosing the method of conserving hay.

- ⇒ Availability of space.
- ⇒ Availability of equipment.
- ⇒ Availability of land.

What is standing forage?



- ✓ Standing forage refers to pasture or forage which is left uncut/ungrazed or still growing undisturbed for either grazing or feeding livestock in the future when it is dry or there is drought.
- ✓ It should be sufficient enough to feed livestock throughout the dry seasons.

Advantages of standing forage.

- Used by livestock when fresh.
- Can be harvested despite the weather available.
- Cheap and easy method of conserving forage.

Disadvantages of standing forage.

- ✗ Requires land.
- ✗ Can be destroyed by weather elements such as hailstones, floods etc.

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Reasons for conserving forage.

- To provide animal feed during dry periods of feed shortage and scarcity.
- To avoid wastage of forage.
- To ensure better and full utilization (use) of available land.
- To get income from selling surplus forage.

Importance of Conserving forage.

- ✓ It ensures stable and continuous supply of pasture for livestock.
- ✓ Reduces the cost of purchasing additional feed for livestock. This help to save.
- ✓ Enhance community commitments to climate change since drought and other extreme weather disasters are less likely to have effects when forage is well managed.

1.2 Conserving of Leftover foods.

- ✓ Leftover food is food that has been prepared and cooked but not all eaten at once.
- ✓ Conserving left over foods refers to keeping them safe for use in order to avoid or not waste it.
- ✓ Left over foods should be conserved to prevent wastage.
- ✓ They can be stored in covered containers or refrigerators to prepare them later or for consumption.

Importance of Conserving leftover foods at home.

Importance of Conserving/Reasons For reusing left over food.

- ✓ To reduce food wastage – instead of throwing away leftover food, it can be transformed into new dishes.
- ✓ To save resources and money – reusing leftovers enable saving of both food and ingredients required to make it. This helps to save money as no more food is bought.
- ✓ To save on labour -no need for the initial food preparations, the leftovers can be used to prepare a new dish.
- ✓ To save time -reheated dishes save time in the kitchen, since cooked leftover do not undergo initial preparation process of cooking again. This saves time.
- ✓ Does not require special skills – since leftover food is already available, preparing it is easier than making a new dish.
- ✓ For food security – using leftovers makes food available for everyone because nothing is wasted. This ensures everyone has enough to eat.
- ✓ Help in keeping environment healthy – reusing leftovers helps the environment as less food is wasted which is good for our environment.

Flour mixtures.

- Flour can be mixed with water and other ingredients to make batters or dough.
- **Flour mixture** refers to a mixture of flour as main agent with water and other ingredients such as a raising agent, sugar, margarine and salt.
- Wheat flour is mainly used to make flour mixtures.
- Wheat contains a substance called gluten which stretches and holds the flour mixture together.
- These flour mixtures are used to make food products such as
 - ⇒ Pancakes.
 - ⇒ Chapati.
 - ⇒ Mandazi.
- The final products from flour mixture depends on how thin or thick the flour mixture is.

Types of Flour mixtures.

There are two types of flour mixtures used to make different types of food products.

They include:

- **Batters.**
- **Doughs.**

a.) Batters.



- These are mixtures made by combining flour with a liquid such as water or milk to form a semi-liquid mixture.
- There are two types of batters namely; thin batters and thick batter.
- Thin batters have running consistency for example for making crepes or pancakes.
- Thick batter has a thick consistency, for example, batter for making cakes.
- Batters are commonly used in the preparation of foods such as:
 - ✓ Pancakes.
 - ✓ Cakes.